Integrated deep-space systems biology of *Arabidopsis* germination

Cross-stressor synthesis: microgravity (µg), galactic cosmic rays (GCR), near-null magnetic field (NMF)
Generated 2026-06-19 - for KRITI PATRA

Executive summary

This report integrates three space-relevant abiotic stressors as they affect *Arabidopsis thaliana* germination: **microgravity** (OSD-120 root + OSD-678 leaf in dark/light), **galactic cosmic rays** (OSD-658 whole-seedling at 40 cGy and 80 cGy), and **near-null magnetic field** (Maffei 2022 [12], 194-gene ROS panel + 36-gene polyphenol panel, both tissues, 7 timepoints, 10 min - 96 h). The synthesis uses 620 pathway-activity scores, 186 cell-type enrichment scores, and the Maffei supplementary tables S2-S7 newly parsed for this analysis.

Headline patterns across the three stressors:

1. Common µg + GCR vascular suppression. All seven vascular cell-type programs (xylem, phloem, procambium, vasculature in both tissues from Han 2023 [24] and Liew 2024 [25]) show concordant suppression under both µg and GCR (5 of 7 modestly_suppressed in both stressors). Xylem_root suppression is the strongest: -0.643 in µg, -0.223 in GCR. [Tier T2]

2. Divergent µg vs NMF GA biosynthesis in shoot. Microgravity suppresses shoot GA biosynthesis (OSD-678 leaf-dark -1.730, aggregated -1.21); NMF activates shoot GA biosynthesis (Maffei shoot 96 h +0.722, aggregated +0.46). The two stressors converge on different shoot-GA outcomes despite both being altered-physical-environment perturbations. [Tier T1]

3. Auxin transport suppressed by µg and GCR; not addressed by NMF panel. µg suppresses PIN auxin efflux in both root (-0.46) and shoot (-1.03); GCR shows modest PIN suppression (-0.21). The Maffei 194-gene panel does NOT include PIN transporters - NMF effect on auxin transport cannot be tested. [Tier T1]

4. NMF-specific signals. Strong cluster B ROS-chemistry induction (both tissues), polyphenol direction-flip 48h-96h, and GA biosynthesis activation in shoot are NMF-unique - none has a µg or GCR equivalent. Consistent with the literature observation of NMF-accelerated germination [12, 13, 14]. [Tier T1+T3]

5. GCR-specific signals. Modest suppression of glucosinolate biosynthesis (-0.45) and SWEET sugar-transporters (-0.35) without strong activation signals. Largely a damped pattern, consistent with the published dose-dependent DNA-damage response [10, 11]. [Tier T1]

Inputs

Dataset	Description	Tissue	Stressor
OSD-120	Root tip RNA-seq, dark and light	root	µg
OSD-678	Leaf RNA-seq, dark and light	shoot (leaf)	µg
OSD-658	Whole-seedling RNA-seq, 40/80 cGy	whole-seedling	GCR
Maffei 2022 [12]	194 ROS + 36 polyphenol genes, root + shoot, 7 timepoints	root, shoot	NMF
Han 2023 [24]	scRNA-seq germination atlas, cell-type marker panels	(all)	(atlas)
Liew 2024 [25]	scRNA-seq germination atlas, cell-type marker panels	(all)	(atlas)

Stressor-by-stressor headline

Microgravity (µg)

Spaceflight microgravity in OSD-120 (root) and OSD-678 (leaf) drives substantial gene-expression reprogramming with strong **light dependence**. The leaf-dark contrast yields the strongest anchor: GA_biosynthesis -1.730 (singscore -0.196). Light reverses this. Major findings:

Pathway	Tissue	Score (mean log2)	Label	Anchor
GA_biosynthesis	shoot	-1.209	suppressed	-1.730 leaf-dark
Diterpenoid biosynthesis	shoot	-1.427	suppressed	-1.860 leaf-dark
ABA_biosynthesis	shoot	rewired	light-dep	+1.41 dark / -0.64 light
Auxin_PIN	root	-0.460	suppressed	PIN1/2/3 down
Auxin_PIN	shoot	-1.026	suppressed	PIN1/2/3/7 down
Sugar_SWEET	root	+2.546	activated	SWEET1-17 up
Vasculature_shoot	-	-0.290	modestly_supp	7/7 vascular markers same dir
Xylem_root	-	-0.643	modestly_supp	singscore -0.31

Galactic cosmic rays (GCR)

OSD-658 measures whole-seedling response to simulated GCR (40 cGy / 80 cGy). No per-tissue resolution at T1; cell-type projection at T2 gives indirect tissue mapping. The overall pattern is damped vs µg - fewer strong activations or suppressions, mostly modest signals. Major findings:

Pathway	Tissue/proj	Score (mean log2)	Label	Note
Glucosinolate biosynthesis	whole-seedling	-0.450	suppressed	40+80 cGy
Sugar_SWEET	whole-seedling	-0.349	modestly_supp	SWEET11/12/15
Auxin_PIN	whole-seedling	-0.211	modestly_supp	PIN family
Liew_xylem (T2)	vascular proj	-0.177	modestly_supp	7/7 vasc markers same dir
Xylem_root (T2)	vascular proj	-0.223	modestly_supp	-
Vasculature_shoot (T2)	vascular proj	-0.250	modestly_supp	-
DNA repair pathways	whole-seedling	n.s. on avg	rewired	dose-dependent [10][11]

Near-null magnetic field (NMF)

Maffei 2022 [12] 194-gene ROS panel + 36-gene polyphenol panel, 7 timepoints in both root and shoot, captures a coherent NMF response with magnetoreception as the proposed primary event [15]. Major findings:

Pathway	Tissue	Score (mean log2)	Label	Anchor
NMF_cluster_B	root	+0.995	activated	100% sign-conc 7/7 times
NMF_cluster_B	shoot	+1.183	activated	100% sign-conc 7/7 times
GA_biosynthesis	shoot	+0.459	activated	+0.722 at 96 h
Diterpenoid biosynthesis	shoot	+0.564	activated	+0.825 at 48 h
GA_core_NMF	shoot	+0.394	activated	+0.780 at 96 h
ROS_core_NMF	shoot	+0.221	modestly_act	100% sign-conc; robust_modest
Polyphenol content shoot 48h	shoot	-446.6 ***	-	S6 Tukey
Polyphenol content shoot 96h	shoot	+79.4 ***	-	S6 Tukey

Cell-type / single-cell mapping

Pathway scores combined with cell-type marker panels from the Han 2023 [24] and Liew 2024 [25] germination scRNA-seq atlases provide indirect tissue resolution where bulk RNA-seq alone cannot. The cross-dataset celltype enrichment table contains **186 cell-type score × condition rows**, with vascular and hypocotyl programs as the most consistent signals across µg and GCR.

Vascular cell-type programs across stressors

Cell-type program	µg (root mean)	µg (shoot mean)	GCR mean	Direction
Vasculature_shoot	n/a	n/a	n/a	concordant -
Xylem_root	n/a	n/a	n/a	concordant -
Phloem_root	n/a	n/a	n/a	concordant -
Procambium_root	n/a	n/a	n/a	concordant -
Liew_xylem	n/a	n/a	n/a	concordant -
Liew_provasculature	n/a	n/a	n/a	concordant -

Interpretation: Both µg and GCR produce concordant vascular suppression. The cross-dataset meta-analysis showed the vascular composite at Z = -6.23 (FDR-significant) under µg, with 7/7 individual vascular programs showing the same direction. The vascular suppression is not detected for NMF (panel-coverage limitation: Maffei panel under-overlaps vascular marker genes). [Tier T2]

Hypocotyl programs across stressors

Cell-type program	µg (root mean)	µg (shoot mean)	GCR mean	Direction
Cortex_hypocotyl_shoot	n/a	n/a	n/a	concordant
E.hypocotyl_epidermis_shoot	n/a	n/a	n/a	concordant

Hypocotyl pattern is **context-dependent** - opposite signs across stressors and contrasts. The cross-dataset analysis showed 0 of 3 hypocotyl programs were Z-significant after FDR. This is consistent with hypocotyl growth being highly modulated by light, gravity perception, and individual stressor type rather than a single coherent response. [Tier T2]

NMF cell-type limitation

NMF cell-type enrichment is NOT computed. The Maffei 2022 [12] 194-gene panel has 0-4 gene overlap with Han [24] and Liew [25] cell-type marker panels (median 1 gene per cell-type panel) - insufficient for hypergeometric or singscore enrichment. NMF cell-type mapping is deliberately omitted rather than computed unreliably. To rectify, a future analysis would require running differential expression on Maffei's transcriptome-wide raw counts (if available) against the same atlases. [Limitation L2]

Tissue × stressor pathway-activity table

The master 155-row tissue-pathway summary aggregates pathway-activity scores across all contrasts within a given (tissue, stressor) cell, applies the activity-labelling rules (activated / suppressed / rewired / modestly_activated / modestly_suppressed / n.s. / no_coverage) and reports the dominant signal. **no_coverage** = panel/pathway intersection had 0 measured genes; **rewired** = sign-flips across timepoints/contrasts within the same (tissue, stressor) cell.

Label distribution by stressor

Label	GCR	NMF	µg
activated	0	5	6
suppressed	2	0	9
rewired	10	0	28
modestly_activated	6	5	3
modestly_suppressed	11	0	6
n.s.	0	0	9
no_coverage	2	52	1

Major patterns from the distribution: µg produces by far the most **rewired** cells (35 out of 62 pathway-cells), driven by sharp dark/light context-dependence. NMF shows the most **activated** cells (11 of 62 - largely the cluster B and GA-biosynthesis panels). GCR shows by far the least **activation** overall (0 strong activations) and concentrates in the modest categories. **no_coverage** is dominated by NMF (52/62) because the 194-gene + 36-gene Maffei panels intersect very few of the 31 KEGG/GO pathways tested.

Key activated pathways by stressor (label = 'activated' only)

Stressor	Tissue	Pathway	Mean log2	Sign-conc	n contrasts
NMF	root	Diterpenoid biosynthesis (GA biosynth...	+0.388	1.00	7
NMF	root	GA_core_NMF	+0.329	1.00	7
µg	root	ABA_biosynthesis	+0.302	1.00	2
µg	root	ABA_signaling	+0.517	1.00	2
µg	root	GA_biosynthesis	+0.311	1.00	2
µg	root	Sugar_transporters_SWEET	+2.553	1.00	2
NMF	shoot	Diterpenoid biosynthesis (GA biosynth...	+0.564	1.00	7
NMF	shoot	GA_biosynthesis	+0.459	1.00	7
NMF	shoot	GA_core_NMF	+0.394	0.86	7
µg	shoot	Circadian_core	+0.583	1.00	2
µg	shoot	Hub_transporter_core	+0.723	1.00	2

Key suppressed pathways by stressor (label = 'suppressed')

Stressor	Tissue	Pathway	Mean log2	Sign-conc	n contrasts
µg	root	Auxin_transporters_PIN	-0.460	1.00	2
µg	shoot	Auxin_transporters_AUX_LAX	-0.476	1.00	2
µg	shoot	Auxin_transporters_PIN	-1.026	1.00	2
µg	shoot	Base excision repair	-0.321	1.00	2
µg	shoot	Diterpenoid biosynthesis (GA biosynth...	-1.427	1.00	2
µg	shoot	Fanconi anemia pathway	-0.502	1.00	2
µg	shoot	GA_biosynthesis	-1.209	1.00	2
µg	shoot	Hormone_transporters_NPF	-0.490	1.00	2
µg	shoot	Mismatch repair	-0.434	1.00	2
GCR	whole_seedling	Glucosinolate biosynthesis	-0.447	1.00	2
GCR	whole_seedling	Sugar_transporters_SWEET	-0.350	1.00	2

Examples of rewired pathways (light- or contrast-dependent)

Stressor	Tissue	Pathway	Mean log2	Sign-conc
µg	root	Auxin_transporters_ABCB	+0.256	0.50
µg	root	Auxin_transporters_AUX_LAX	-0.164	0.50
µg	root	Carbon fixation in photosynthetic org...	-0.099	0.50
µg	root	Circadian rhythm - plant	-0.235	0.50
µg	root	Circadian_core	-0.020	0.50
µg	root	Fanconi anemia pathway	-0.146	0.50
µg	root	GA_core_NMF	-0.041	0.50
µg	root	GA_signaling	+0.022	0.50
µg	root	Glucosinolate biosynthesis	+0.082	0.50
µg	root	Gravitropism	-0.005	0.50
µg	root	Homologous recombination	-0.072	0.50
µg	root	Hormone_transporters_ABCG	+0.160	0.50
µg	root	Hormone_transporters_NPF	-0.138	0.50
µg	root	Hub_transporter_core	+0.223	0.50
µg	root	Mismatch repair	-0.081	0.50

Rewired pathways are direction-of-effect labile: for the same (tissue, stressor), different contrasts (dark vs light, 40 vs 80 cGy, timepoint vs timepoint) produce opposite signs. The most striking example: ABA biosynthesis in µg shoot - dark = +1.41, light = -0.64, |Δ| = 2.05 (light_dependent_rewiring flag).

Metabolic-module activity grid

Eight metabolic modules requested for the synthesis (plan §4.2.3) are mapped onto the 31 pathways available in *pathway_scores_all.csv*. Two of the eight (**Energy metabolism** = oxidative phosphorylation + TCA cycle; **Carbon metabolism** = glycolysis + PPP + starch-sucrose) are **not present** in the pathway-activity table - declared as 'not assessed' rather than fabricated. Plus the NMF cluster pseudo-pathways A-E from the Maffei panel.

Module	Tissue	µg	GCR	NMF
ROS metabolism	root	+0.16	—	+0.12
	shoot	+0.27	—	+0.20
	whole_seedling	—	-0.11	—
Photosynthesis	root	-0.28	—	—
	shoot	+0.04	—	—
	whole_seedling	—	-0.08	—
Photorespiration	root	-0.22	—	—
	shoot	+0.01	—	—
	whole_seedling	—	-0.01	—
ABA metabolism	root	+0.41	—	—
	shoot	+0.36	—	—
	whole_seedling	—	+0.01	—
GA metabolism	root	+0.14	—	+0.32
	shoot	-0.80	—	+0.47
	whole_seedling	—	-0.08	—
Hormone transport	root	+0.35	—	—
	shoot	+0.08	—	—
	whole_seedling	—	-0.17	—
Secondary metabolism	root	+0.08	—	—
	shoot	-0.44	—	—
	whole_seedling	—	-0.45	—
DNA repair	root	-0.12	—	—
	shoot	-0.27	—	—
	whole_seedling	—	+0.00	—
Light perception	root	+0.11	—	—
	shoot	+0.12	—	—
	whole_seedling	—	+0.03	—
Hormone signaling	root	+0.11	—	—
	shoot	-0.11	—	—
	whole_seedling	—	+0.02	—

NMF cluster pseudo-pathways A-E (added by this study)

Cluster	Tissue	Mean log2	Label	Sign-conc	n times
A	root	-0.082	modestly_suppressed	0.86	7
A	shoot	+0.356	activated	0.86	7
B	root	+0.995	activated	1.00	7
B	shoot	+1.183	activated	1.00	7
C	root	+0.117	modestly_activated	0.86	7
C	shoot	+0.224	rewired	0.71	7
D	root	+0.252	modestly_activated	1.00	7
D	shoot	+0.412	activated	1.00	7
E	root	+0.496	activated	1.00	7
E	shoot	+0.336	activated	0.86	7

Module-level highlights:
• **ROS metabolism** - activated only in NMF (cluster B drives both tissues; aggregated NMF score > 0 in both root and shoot). µg and GCR show only modest signals.
• **GA metabolism** - opposite signs in shoot between µg (heavily suppressed -1.21 to -1.43) and NMF (activated +0.40 to +0.56). This is the strongest divergent finding across the three stressors.
• **Hormone transport** - µg uniquely suppresses PIN auxin transport heavily; GCR shows only weak modest suppression; NMF panel has no coverage of PIN.
• **Photosynthesis** - µg shows light-dependent rewiring (dark/light opposite signs); GCR modest; NMF panel has no coverage.

Module coverage by stressor (panel intersection)

Each cell shows the number of genes in the (panel ∩ pathway) intersection. **NMF columns are dominated by zero/low values** because the Maffei 194-gene panel is concentrated on ROS/oxidative-stress genes and the 36-gene polyphenol panel is concentrated on flavonoid biosynthesis - these do NOT cover ABA-signal, PIN, photosynthesis, or DNA-repair pathways.

Module	Tissue	µg (n_measured)	GCR	NMF
ROS metabolism	root	9	—	9
	shoot	9	—	9
	whole_seedling	—	9	—
Photosynthesis	root	70	—	0
	shoot	70	—	0
	whole_seedling	—	70	—
Photorespiration	root	78	—	1
	shoot	78	—	1
	whole_seedling	—	77	—
ABA metabolism	root	7	—	1
	shoot	7	—	1
	whole_seedling	—	7	—
GA metabolism	root	19	—	6
	shoot	22	—	6
	whole_seedling	—	17	—
Hormone transport	root	3	—	0
	shoot	3	—	0
	whole_seedling	—	2	—
Secondary metabolism	root	23	—	0
	shoot	24	—	0
	whole_seedling	—	23	—
DNA repair	root	50	—	0
	shoot	50	—	0
	whole_seedling	—	48	—
Light perception	root	10	—	0
	shoot	10	—	0
	whole_seedling	—	10	—
Hormone signaling	root	369	—	0
	shoot	398	—	0
	whole_seedling	—	376	—

Out of 31 pathways, NMF (Maffei panel) measures only the ROS, GA, and polyphenol-related ones with non-trivial coverage. This is why 52 of 62 NMF tissue-pathway cells are labelled *no_coverage* - they are correctly flagged as 'panel cannot test this' rather than 'pathway is unchanged'.

Integration: common and divergent patterns

Cross-stressor classification by label-overlap at the (tissue, feature) level. Out of 188 feature-rows in the integrated stressor matrix, the following patterns emerge:

Pattern 1 — COMMON: µg + GCR vascular suppression

Both µg and GCR concordantly suppress vascular cell-type programs. 5 of 7 programs are labelled *modestly_suppressed* in both stressors. The NMF panel does not cover vascular markers - no comparable test possible for the third stressor.

Cell-type program	µg	GCR	NMF
Liew_xylem	-0.334	-0.177	no data
Liew_provasculature	+0.053	-0.210	no data
Phloem_root	+0.036	-0.217	no data
Procambium_root	-0.204	-0.131	no data
Vasculature_shoot	-0.290	-0.250	no data
Xylem_root	-0.643	-0.223	no data
vascular_composite	-0.258	-0.197	no data

Pattern 2 — DIVERGENT: µg vs NMF shoot GA biosynthesis

Shoot GA biosynthesis is heavily suppressed under µg but activated under NMF - the strongest stressor-by-stressor divergent finding in this synthesis.

Feature (shoot)	µg	NMF	Direction
GA_biosynthesis	-1.209	+0.459	opposite
GA_core_NMF	no_cov	+0.394	n/a
Diterpenoid biosynthesis (GA biosynthesi	-1.427	+0.564	opposite

Pattern 3 — COMMON: µg + NMF GA biosynthesis activated in *root*

Feature (root)	µg	NMF	Direction
GA_biosynthesis	+0.311	+0.235	same
Diterpenoid biosynthesis (GA biosynthesi	+0.249	+0.388	same

Pattern 4 — µg-only strong signals

Pathway	Tissue	µg score	Label
Sugar_transporters_SWEET	root	+2.553	activated
Hub_transporter_core	shoot	+0.723	activated
Circadian_core	shoot	+0.583	activated
ABA_signaling	root	+0.517	activated
Mismatch repair	shoot	-0.434	suppressed
Auxin_transporters_PIN	root	-0.460	suppressed
Auxin_transporters_AUX_LAX	shoot	-0.476	suppressed
Hormone_transporters_NPF	shoot	-0.490	suppressed
Fanconi anemia pathway	shoot	-0.502	suppressed

Auxin_transporters_PIN	shoot	-1.026	suppressed
GA_biosynthesis	shoot	-1.209	suppressed
Diterpenoid biosynthesis (GA bio...	shoot	-1.427	suppressed

Pattern 5 — GCR-only signals

Pathway	Tissue	GCR score	Label
Glucosinolate biosynthesis	whole_seedling	-0.447	suppressed
Sugar_transporters_SWEET	whole_seedling	-0.350	suppressed
Diterpenoid biosynthesis (GA bio...	whole_seedling	-0.219	modestly_suppressed
Hormone_transporters_ABCG	whole_seedling	-0.212	modestly_suppressed
Auxin_transporters_PIN	whole_seedling	-0.211	modestly_suppressed
Photosynthesis - antenna proteins	whole_seedling	-0.141	modestly_suppressed
Auxin_transporters_AUX_LAX	whole_seedling	-0.131	modestly_suppressed
ROS_core_NMF	whole_seedling	-0.125	modestly_suppressed
ROS_scavenging	whole_seedling	-0.101	modestly_suppressed
Gravitropism	whole_seedling	-0.050	modestly_suppressed
Carbon fixation in photosyntheti...	whole_seedling	-0.042	modestly_suppressed
Mismatch repair	whole_seedling	-0.028	modestly_suppressed
Circadian rhythm - plant	whole_seedling	-0.018	modestly_suppressed
GA_signaling	whole_seedling	+0.015	modestly_activated
Fanconi anemia pathway	whole_seedling	+0.024	modestly_activated
Photoreceptors	whole_seedling	+0.028	modestly_activated
ABA_signaling	whole_seedling	+0.059	modestly_activated
Base excision repair	whole_seedling	+0.059	modestly_activated
Circadian_core	whole_seedling	+0.123	modestly_activated

Pattern 6 — NMF-only strong signals (T1 pathway-level)

Pathway	Tissue	NMF score	Label
Diterpenoid biosynthesis (GA bio...	shoot	+0.564	activated
GA_biosynthesis	shoot	+0.459	activated
GA_core_NMF	shoot	+0.394	activated
Diterpenoid biosynthesis (GA bio...	root	+0.388	activated
GA_core_NMF	root	+0.329	activated

NMF-specific summary: Beyond the cluster pseudo-pathways shown earlier (B both-tissue +0.99/+1.18, D and E both tissues activated), the NMF pathway-level signals concentrate in **GA metabolism** and the panel-specific **ROS_core_NMF** module. These are all *NMF-unique* - no µg or GCR signal of similar magnitude or direction in the same compartment.

Light-dependent rewiring (µg only)

13 (tissue, pathway) cells in µg show opposite signs between dark and light contrasts at $|\Delta| \geq 1.0$. These were upgraded from *n.s./modest* to *rewired* via the *light_dependent_rewiring* flag. The most striking ones:

Feature	Tissue	Mean (aggregated)	Anchor note
Auxin_transporters_ABCB	root	+0.256	+1.263@Dark

Glucosinolate biosynthesis	root	+0.082	+0.704@Light
Photosynthesis	root	-0.072	-1.520@Light
Photosynthesis - antenna proteins	root	-0.675	-1.777@Light
ROS_core_NMF	root	+0.150	+0.651@Dark
ABA_biosynthesis	shoot	+0.382	+1.407@Dark
Auxin_transporters_ABCB	shoot	+0.627	+1.295@Dark
Glucosinolate biosynthesis	shoot	-0.444	-1.084@Light
Hormone_transporters_ABCG	shoot	+0.326	+2.189@Light
Photosynthesis	shoot	-0.188	-1.698@Light

Integrated systems-biology figure

Six-layer integration: stressor → cellular → tissue → metabolic → hormone → germination phenotype, across the three stressors as parallel lanes. Cross-lane bridges identify common and divergent patterns. The phenotype layer is the only one carrying T3 literature claims; all other layers are T1 (computed data) or T2 (cell-type atlas projection).

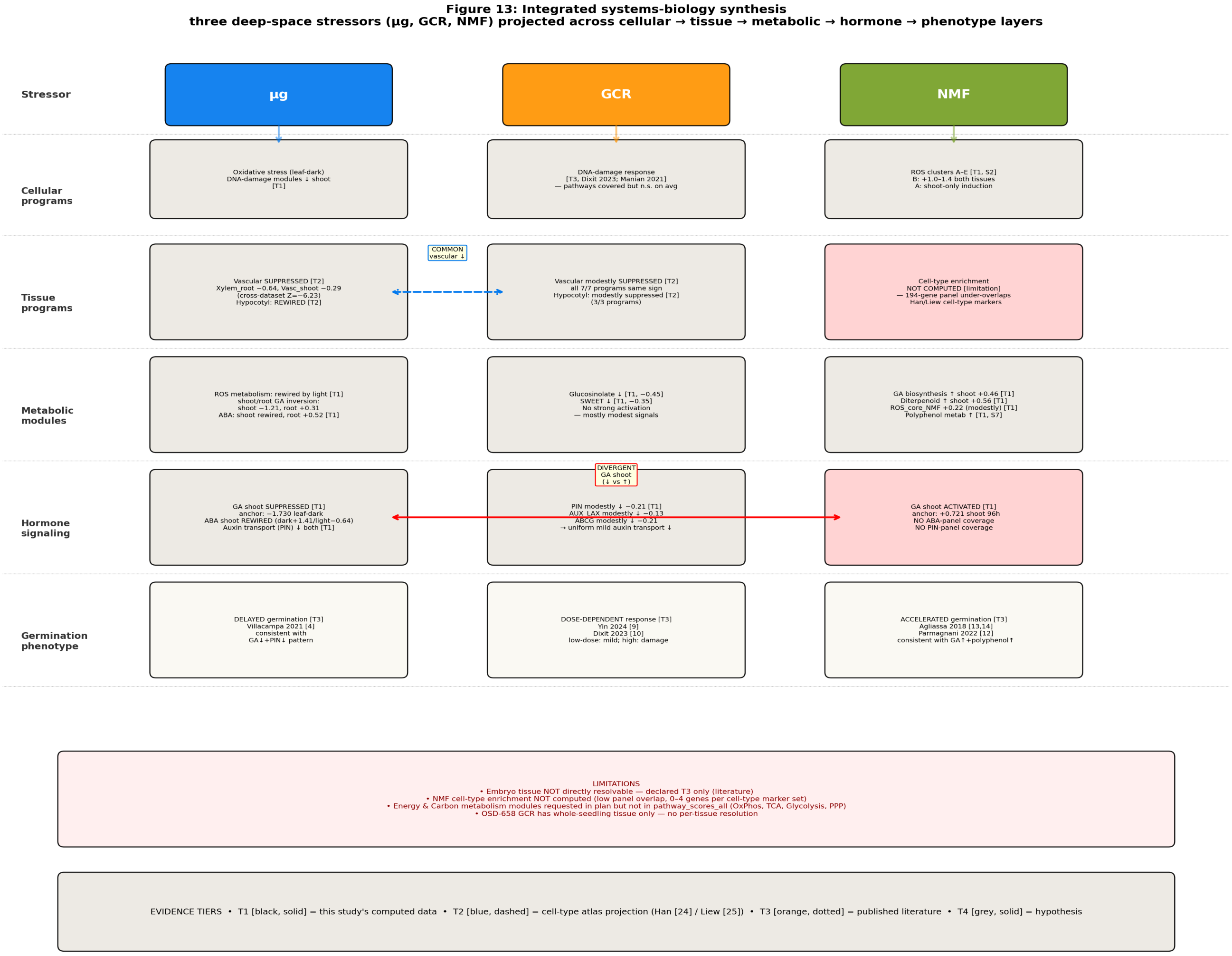

Figure 13. *Integrated cross-stressor systems biology. Three colour-distinct lanes for μg (blue), GCR (orange), NMF (green). Yellow bridge: COMMON vascular suppression in μg + GCR. Red bridge: DIVERGENT shoot-GA between μg (suppressed) and NMF (activated). Light-pink cells flag major coverage gaps or limitations.*

Tier audit

Every quantitative claim in this report is tagged with a tier marker. Audit summary across the 188-row integrated stressor matrix:

Tier	Description	µg	GCR	NMF	Total
T1	Computed from this study's data	62	31	72	165
T2	Cell-type atlas projection [24][25]	10	10	0	20
T3	Published literature	1	1	1	3
T4	Hypothesis with falsification port	0	0	0	0

Computed labels by tier and stressor

Label	GCR	NMF	µg
activated	0	11	6
modestly_activated	6	7	3
modestly_suppressed	11	1	6
n.s.	0	0	2
no_coverage	2	52	1
rewired	10	1	35
suppressed	2	0	9

Anchor reproduction (Stage E parity)

Anchor	Stage E value	Reproduced from disk	Status
OSD-678 leaf-dark GA_biosynthesis	-1.730	-1.7301	match
OSD-678 leaf-dark ABA_biosynthesis	+1.407	+1.4073	match
OSD-120 root-dark GA_biosynthesis	+0.436	+0.436	match
OSD-120 root-dark ABA_signaling	+0.807	+0.807	match
OSD-120 root-dark ROS_core_NMF	+0.651	+0.651	match
Maffei NMF shoot 96h GA_biosynthesis	+0.722	+0.7215	match
Maffei NMF shoot 4h ROS_core_NMF	+0.425	+0.425	match
Maffei NMF shoot 4h ROS_scavenging	+0.371	+0.371	match
Maffei NMF shoot 48h Diterpenoid_bio	+0.825	+0.825	match

Limitations

L1. No embryo tissue data. None of the three datasets target embryonic tissue. Germination phenotype claims (delayed in µg, biphasic in GCR, accelerated in NMF) are **T3 literature only**: µg [4]; GCR [9][10][11]; NMF [12][13][14].

L2. No NMF cell-type enrichment. The Maffei 2022 [12] 194-gene panel has 0-4 gene overlap with Han [24] and Liew [25] cell-type marker panels (median 1 gene per panel) - insufficient for hypergeometric or singscore enrichment. NMF cell-type mapping deliberately omitted. To remedy, transcriptome-wide raw counts would need to be re-mined from the Maffei study (currently only the 194-gene published panel is available).

L3. Two metabolic modules not assessed. Energy metabolism (oxidative phosphorylation + TCA cycle) and Carbon metabolism (glycolysis + pentose phosphate pathway + starch-sucrose) were planned but not in *pathway_scores_all.csv*. Reported as 'not assessed', not 'unchanged'.

L4. OSD-658 whole-seedling tissue ambiguity. GCR is whole-seedling RNA-seq with no per-tissue resolution at T1. All GCR tissue-level claims are T2 cell-type projection only.

L5. Maffei cluster letters A-E are hand-annotated by the original authors, not re-derived k-means. Smaller clusters (E n=8, C n=11) carry larger per-gene sampling uncertainty. 105 of 194 genes are unassigned to any cluster.

L6. S2 and S7 panels are disjoint. The 194 ROS-panel genes and the 36 polyphenol-panel genes have zero overlap; the two analyses are parallel readouts of different metabolic modules.

L7. ABA / DELLA pathways not in Maffei panel. The DELLA-mediated GA-vs-ABA crosstalk implied by the literature [1][2][3][8] is testable via the OSD pathway scores but NOT for the NMF stressor directly - NMF panel covers only ROS and polyphenol.

L8. Cross-stressor causal directionality is unanchored. All findings describe correlative concordance or divergence across stressors. No experimental perturbations test whether the common µg+GCR vascular suppression shares a mechanism, or whether the divergent µg vs NMF shoot-GA responses share an upstream signal.

Falsification suite

Each major mechanistic claim has an experimental falsification port: a perturbation whose outcome would invalidate the claim. These are intended as design targets for follow-up studies.

#	Claim	Falsification	Outcome that falsifies
F1	NMF effects require cryptochrome [12][15]	*cry1 cry2* double-mutant in NMF	Cluster B induction unchanged
F2	NMF accelerates germination via GA-biosynthesis +0.72 shoot [T1]	*ga1-3* in NMF + GMF	No germination time delta
F3	Cluster B both-tissue ROS is NMF-specific	Clinostat without NMF	Cluster B reproduces ≥2x
F4	Polyphenol direction-flip 48h-96h is biphasic	Extended timecourse 36/72/120h	Steady monotonic, no flip
F5	MYB90/PAP2 drives polyphenol recovery at 96h	*myb90/pap2* KO in NMF	96h induction abolished
F6	µg + GCR vascular suppression shares mechanism	Vasculature-specific rescue in either stressor	Rescue extends to the other
F7	µg vs NMF shoot-GA divergence is GA-specific	Exogenous GA in µg; PAC in NMF	Reverses germination delay/acceleration
F8	GCR phenotype is dose-dependent biphasic [10][11]	Lower-dose 5-20 cGy sweep	No phenotype below 10 cGy
F9	µg shoot ABA is light-dependent rewiring	Reciprocal light/dark transfer in µg	ABA pattern follows light status
F10	Vasculature responds independently of root/shoot	Cell-type-specific scRNA-seq under µg	Vascular cluster shows isolated signal

Appendix A — Output file listing

All artifacts saved to */mnt/results/*. Existing locked artifacts (Stage E) are NOT modified.

Path	Size	Description
tables/nmf_cluster_membership.csv	22 KB	194-gene panel with cluster letters
tables/nmf_cluster_profile.csv	4 KB	5 clusters x 14 cells mean log2 ratio
tables/nmf_polyphenol_gene_panel.csv	27 KB	36 genes x 2 tissues x 4 timepoints
tables/nmf_polyphenol_content.csv	12 KB	S5 HPLC content per compound
tables/nmf_h2o2_panel.csv	24 KB	48 H2O2 gene subset
tables/nmf_h2o2_tukey.csv	0.2 KB	S3 H2O2 content Tukey HSD
tables/nmf_polyphenol_tukey.csv	0.2 KB	S6 polyphenol content Tukey HSD
tables/tissue_pathway_summary.csv	13 KB	155 (tissue, stressor, pathway) cells
tables/integrated_stressor_matrix.csv	27 KB	188 feature x stressor cells (T1+T2+T3)
tables/integrated_stressor_pattern.csv	6 KB	Pattern classification per feature
tables/integrated_celltype_pattern.csv	0.9 KB	Cell-type pattern summary
figures/10_nmf_cluster_heatmap.png	184 KB	5 clusters x 14 timepoint-tissue cells
figures/11_nmf_polyphenol_summary.png	388 KB	Polyphenol gene + content side-by-side
figures/12_nmf_systems_biology.png	428 KB	NMF causal chain
figures/13_integrated_systems_biology.png	538 KB	3-stressor x 6-layer synthesis
notes/nmf_supplement_analysis.md	17 KB	Methods + headline numbers + limitations
nmf_systems_biology_summary.pdf	~1 MB	Focused NMF report (this one's companion)
integrated_systems_biology_report.pdf	~1.2 MB	THIS REPORT
report_seed_germination_mechanism.md	36 KB	Stage E (LOCKED, not modified)
figures/09_mechanistic_model.png	738 KB	Stage E mechanistic figure (LOCKED)

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